Project 1: Concept Bottleneck Models for Facial Attribute Analysis

Carla Aranda Sánchez: 100523031
Jorge Barcia Belinchón: 100496595
Marina Juzgado Gómez-Menor: 100523023
Iván López Anca: 100523112

***Abstract*—This project evaluates the trade-offs between predictive performance, interpretability, and steerability in Concept Bottleneck Models (CBM). Using the CelebA dataset and a ResNet-18 backbone, we implement and compare four model variants: a baseline classifier, a concept predictor, a pure CBM, and a hybrid architecture with a direct side channel. We investigate the impact of side-channel dropout on model reliance and perform intervention analysis to rank concept influence on the final "Smiling" target. Since several concept labels are imbalanced, we train with `pos_weight` to improve concept learning. Overall, the models achieve strong test performance, with the baseline reaching 0.9323 accuracy and 0.9834 AUROC, while the pure CBM exhibits a drop in target performance and concept quality compared to the standalone concept predictor.**

***Index Terms*—Concept Bottleneck Models, Interpretability, Steerability, ResNet-18, CelebA, Side-channel Dropout.**

I. Introduction

TRADITIONAL deep learning models often function as black boxes, providing high accuracy at the cost of transparency. Concept Bottleneck Models (CBM) aim to bridge this gap by forcing the model to predict human-understandable concepts before reaching a final decision. This report explores these architectures using the CelebA dataset, focusing on the "Smiling" prediction target and a predefined set of 10 facial attribute concepts.

II. Experimental Setup

A. Dataset and Pre-processing

We utilize the "Heavy version" of the project using the **CelebA** dataset. The target label is *Smiling*, and the concept set includes 10 attributes: *Mouth Slightly Open, High Cheekbones, Chubby, Narrow Eyes, Bags Under Eyes, Big Lips, Big Nose, Pointy Nose, Bushy Eyebrows,* and *Arched Eyebrows*. Images are resized to 128×128 and normalized according to ImageNet statistics. CelebA attributes are provided in $\{-1,+1\}$ and were mapped to $\{0,1\}$ prior to training.

Since several concept labels are markedly imbalanced in the training split, we use Binary Cross-Entropy with `pos_weight` computed from the training data. Under class imbalance, a standard loss may be dominated by the majority negative class and lead to weaker learning of rare positive instances. The `pos_weight` term increases the contribution of positive samples to the loss, helping the model learn more faithful concept representations. This is especially important in our setting, since concept quality is itself a central objective of Concept Bottleneck Models.

B. Architectural Backbone

Following the project instructions, we use a **ResNet-18** initialized with ImageNet-pretrained weights. The final classification layer is removed, providing a shared 512-dimensional feature backbone for all variants.

C. Training Details

All models were trained using Adam with learning rate 10^{-4} for 3 epochs. We used the official CelebA train, validation, and test splits. The best checkpoint for each model was selected according to validation loss and then evaluated on the test set. Binary Cross-Entropy with logits was used for both the target and concept losses, with `pos_weight` computed from the training split to account for class imbalance.

III. Models Implementation

A. Baseline and Concept Predictor (M1 & M2)

- **Baseline ($\mathbf{x} \rightarrow \mathbf{y}$):** A standard classifier mapping backbone features directly to the target.
- **Concept Predictor ($\mathbf{x} \rightarrow \mathbf{c}$):** A multi-label head predicting the 10-bit concept vector using Binary Cross-Entropy loss.

B. Concept Bottleneck Model (M3)

We implemented a **Joint CBM** (x $\rightarrow$ c $\rightarrow$ y) trained end-to-end. The label head operates strictly on the predicted concepts, ensuring that the decision path is fully interpretable.

C. Hybrid CBM with Side Channel (M4)

The hybrid model combines the concept path $f(c)$ and a direct side channel $s(x)$: $y = f(c) + s(x)$. To study the reliance on concepts, we apply **dropout** $p \in \{0.0, 0.1, 0.3, 0.5, 0.7, 0.9\}$ exclusively to the side-channel head during training.

IV. Quantitative Results

A. Dropout Sweep Analysis

The side-channel dropout experiment reveals how the model shifts reliance from raw features to concepts. Table III reports test results for the hybrid model (M4) across dropout rates.

TABLE I
BASELINE (M1) VS PURE CBM (M3) PERFORMANCE (TEST SET)

Model	Test Accuracy	Test AUROC	Mean Concept F1
M1 (Baseline)	0.9323	0.9834	–
M3 (Joint CBM)	0.9146	0.9691	0.5860

TABLE II
CONCEPT PREDICTOR (M2) TEST PERFORMANCE (F1 PER CONCEPT)

Concept	Test F1
Mouth_Slightly_Open	0.9328
High_Cheekbones	0.8681
Chubby	0.4050
Narrow_Eyes	0.5084
Bags_Under_Eyes	0.6290
Big_Lips	0.5803
Big_Nose	0.6108
Pointy_Nose	0.5945
Bushy_Eyebrows	0.6493
Arched_Eyebrows	0.7236
Macro-F1 (mean)	**0.6502**

TABLE III
HYBRID CBM (M4) DROPOUT SWEEP RESULTS (TEST SET)

Dropout p	Accuracy	AUROC	Concept Macro-F1
0.0	0.9316	0.9827	0.6555
0.1	0.9275	0.9826	0.6503
0.3	0.9279	0.9822	0.6480
0.5	0.9317	0.9822	0.6499
0.7	0.9240	0.9829	0.6597
0.9	0.9292	0.9832	0.6488

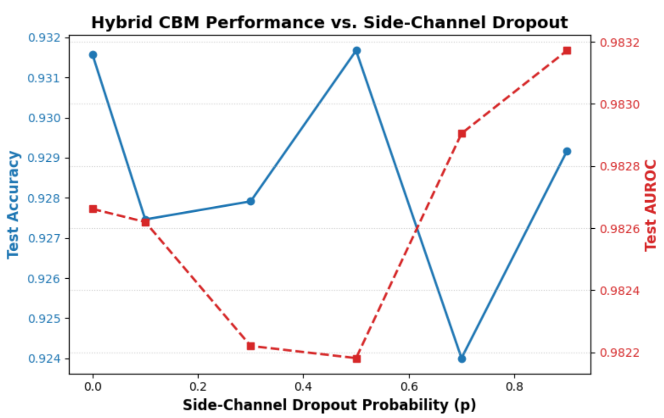

Fig. 1. Test Accuracy and AUROC of the hybrid model (M4) as a function of side-channel dropout p.

V. STEERABILITY AND INTERVENTIONS

To evaluate controllability, we performed concept-level interventions on the test set using the hybrid model with high dropout ($p = 0.9$). We performed continuous soft-interventions on the predicted concept probabilities. For each concept, we inverted its predicted probability p by computing $(1.0 - p)$, kept all other concepts fixed, and recomputed the "Smiling" probability. We report the mean absolute probability change and the fraction of samples whose predicted label flips.

TABLE IV
CONCEPT INTERVENTION RANKING (STEERABILITY) ON M4 WITH $p = 0.9$

Rank	Concept	Avg. Prob. Change	Label Flip Rate (%)
1	Bushy_Eyebrows	0.0095	1.04
2	Narrow_Eyes	0.0066	0.78
3	Bags_Under_Eyes	0.0058	0.71
4	Big_Lips	0.0058	0.65
5	Mouth_Slightly_Open	0.0052	0.59
6	High_Cheekbones	0.0023	0.31
7	Arched_Eyebrows	0.0022	0.26
8	Chubby	0.0019	0.26
9	Big_Nose	0.0012	0.15
10	Pointy_Nose	0.0011	0.15

VI. CRITICAL DISCUSSION

A. Accuracy vs. Interpretability

The Baseline (M1) achieves the best overall target performance (0.9323 accuracy, 0.9834 AUROC) but offers no explicit interpretability. The Pure CBM (M3) achieves lower target performance (0.9146 accuracy, 0.9691 AUROC), while enforcing a fully concept-based decision pathway. However, the concept prediction quality within the pure CBM is lower (macro-F1 0.5860) than the standalone concept predictor (macro-F1 0.6502), illustrating the performance cost of strictly bottlenecked reasoning.

B. Accuracy vs. Steerability

There is a clear trade-off between predictive accuracy and steerability. If the model relies completely on direct visual features, it achieves high accuracy, but it cannot be controlled through concepts. In our Hybrid model (M4), we observed that using a high side-channel dropout ($p = 0.9$) forces the network to actually use the concept bottleneck. This makes the model steerable, as shown by the label flip rates during our interventions (e.g., 1.04% for *Bushy_Eyebrows*). Therefore, to gain steerability and be able to control the model's decisions, we have to penalize the direct pathway, which naturally results in a slight drop in accuracy compared to the baseline.

C. Effect of Side-channel Dropout

Across the dropout sweep, M4 maintains stable target performance around 0.92–0.93 accuracy and ≈0.982–0.983 AUROC (Table III), while consistently achieving concept macro-F1 values around 0.65. This suggests that moderate-to-high side-channel regularization does not substantially degrade predictive accuracy on this task. In addition, the use of `pos_weight` helps improve concept learning under class imbalance, especially in M2 and M4. From a methodological perspective, this weighted setting is justified because it reduces the tendency of the loss to disproportionately favor the majority class, which is particularly important when concept fidelity is one of the main criteria of interest. At high dropout ($p = 0.9$), the intervention analysis shows which concepts produce the largest changes in the final prediction, with `Bushy_Eyebrows`, `Narrow_Eyes`, and `Bags_Under_Eyes` ranking highest in this setting.

VII. Conclusion

The experiments show that concept-based models can approach baseline predictive performance while enabling transparent intermediate representations and post-hoc steerability analysis. The hybrid CBM provides a practical compromise: it preserves strong target performance while allowing concept-level interventions that reveal which attributes most influence "Smiling" predictions. The use of `pos_weight` is justified by the class imbalance of several concepts and improves concept-level performance, particularly in M2 and M4, although the pure CBM pays a noticeable price in target-task accuracy under this weighted setting. Therefore, the weighted setting is not only a practical choice for handling class imbalance, but also a principled one given that accurate concept prediction is a core requirement of the CBM framework.